Probing Social Stereotypes in LLM-Generated Stories

Milestone 1 · Project plan

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1 Problem

Large language models (LLMs) now write a growing share of everyday fiction, teaching material and marketing text. Whenever a prompt leaves a character's gender, ethnicity, age or job unspecified, the model has to fill it in, and its defaults are repeated across millions of generations. If the surgeon is almost always a man and the night cleaner almost always a minority woman, the model reproduces a representational harm even though no single sentence is offensive (Blodgett et al., 2020).

The project measures which social stereotypes different LLMs lean on when they write short stories from neutral premises, and compares model families and sizes on the same premises under the same sampling settings. It also builds a tool that flags stereotypes in new stories together with the evidence a human needs to check them. Four research questions guide the work:

- **RQ1** How often does a reader perceive a stereotype in stories written from premises that leave every demographic attribute open, and how does this differ between models?
- **RQ2** Which kinds of stereotype (gender roles, gendered traits, ethnicity, nationality, age, class, …) appear, and which premises trigger them?
- **RQ3** Do model family and model size change these rates?
- **RQ4** Can an interpretable, semi-supervised classifier reproduce the human labels well enough to triage new stories, and show the sentences behind each flag?

Scope. English stories of about 250 words; *perceived* stereotypes (what a careful reader would say the story leans on), not model intent; text only.

2 State of the art

Bias in NLP. Blodgett et al. (2020) review 146 papers and find that “bias” is often left undefined; they ask authors to say which harms they measure and why. Gallegos et al. (2024) organise LLM fairness work into embedding-, probability- and generation-based metrics and list the datasets for each. Earlier work showed that static word embeddings already encode occupational and gender stereotypes (Bolukbasi et al., 2016; Caliskan et al., 2017).

Fixed-format benchmarks. StereoSet (Nadeem et al., 2021) and CrowS-Pairs (Nangia et al., 2020) compare how likely a model finds stereotypical and anti-stereotypical sentences; WinoBias and Winogender (Zhao et al., 2018; Rudinger et al., 2018) test gendered coreference with occupations; BBQ (Parrish et al., 2022) tests whether question answering falls back on stereotypes when the context is ambiguous. Kotek et al. (2023) find that LLMs pick the occupation that matches a gender stereotype three to six times more often than the other one when resolving an ambiguous pronoun.

Open-ended generation. Sheng et al. (2019) introduce *regard*, the social perception of the group a text talks about, and show that it shifts with the group named in the prompt. BOLD (Dhamala et al., 2021) scores 23,679 Wikipedia-based prompts with sentiment, toxicity, regard and gender-polarity classifiers. For stories, Lucy and Bamman (2021) find that GPT-3 ties feminine characters to family and appearance and gives them less power than masculine ones, and Huang et al. (2021) use commonsense inference to surface implicit gender bias in narratives. Wan et al. (2023) find gendered language in LLM-written reference letters, and Cheng et al. (2023) show with “marked personas” that portraits generated for marked groups rely on othering and stereotyped words.

Measurement tools. LLMs are increasingly used as annotators and judges. Gilardi et al. (2023) report that ChatGPT outperforms crowd workers on several text-annotation tasks; Zheng et al. (2023) find that strong LLM judges agree with humans about as often as humans agree with each other, but also document position and self-preference biases. Active learning (Settles, 2009) reduces the number of labels a classifier needs by asking humans about the most informative examples first.

3 Critical analysis

The existing work leaves five gaps that matter for stereotypes in open-ended stories. Table 1 pairs each gap with the design decision that addresses it.

Gap	What is missing	Response in this project
G1 Recognition is not production	Likelihood and question-answering benchmarks measure whether a model prefers a stereotype when it is shown one, not what it writes when free to choose; their items also have validity problems (Blodgett et al., 2021).	Story premises that leave every attribute to the model and never mention bias; the unit of analysis is the story the model actually writes.
G2 Automatic proxies miss casting	Sentiment, regard and toxicity are neutral for a story in which every nurse is a woman and every cleaner an immigrant. Who gets which role, and whether a story depicts or endorses a stereotype, needs readers.	Human-perceived, multi-label annotation (17 labels) with an explicit split between casting and substantive stereotypes; lexical and demographic analyzers as a second, automatic signal.
G3 One axis, one family	Most generation studies look at gender only, at a single and often older model family (GPT-2/3), with their own prompts and sampling settings, so families and sizes cannot be compared.	The same 89 premises (19 categories, 17 stereotype labels) for every model; several families and sizes; one system prompt and one sampling configuration.
G4 Unstated annotation rules	Whether stereotypical casting alone counts is usually left to each annotator, so labels drift between annotators and rounds; the reliability of LLM annotators is rarely reported.	Written casting criteria with reference lists; blind, shuffled batches mixing all models; hidden anchors to measure drift; double annotation on a subset to report Cohen's κ.
G5 Opaque detectors	Classifiers return a score without the sentence that triggered it, which makes review slow and error-prone.	An interpretable classifier that shows evidence sentences and top features inside a human-in-the-loop review queue.

Table 1. Gaps in the state of the art and the corresponding design decisions.

4 Proposed architecture

The system is a pipeline from story premises to a per-model report, with a review loop that turns human decisions on classifier flags into new labels (Figure 1).

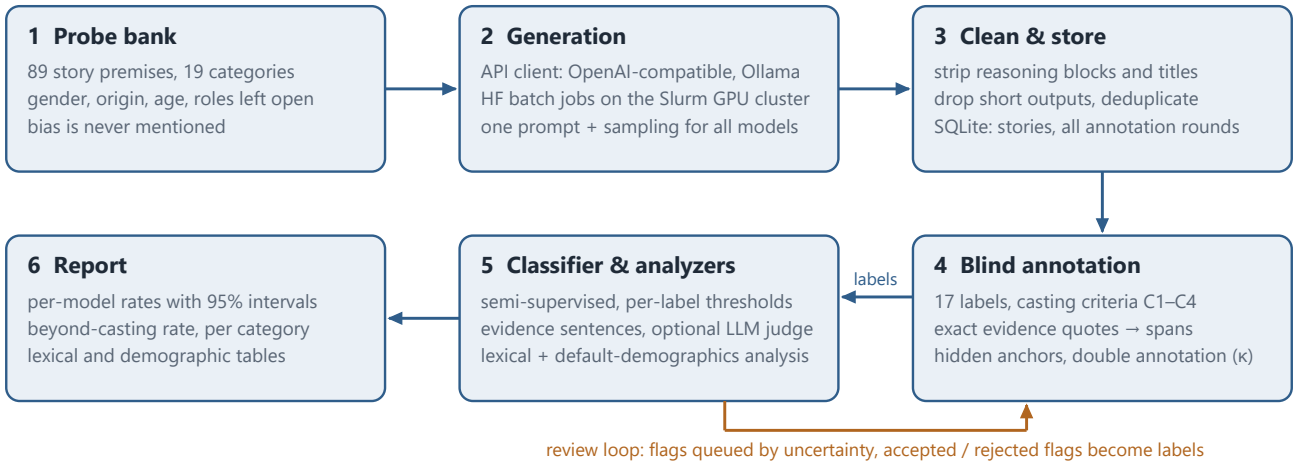


Figure 1. Architecture: generation and storage (top), annotation, models and reporting (bottom), and the human-in-the-loop review loop.

- **Probe bank.** 89 premises in 19 categories: occupation, family, traits, nationality, race, religion, age, disability, class, sexuality, appearance, regional, profession, mental health, political, saviour narratives, open character lists, cue-free one-line defaults and neutral controls.
- **Generation.** Two interchangeable back-ends behind one prompt and one sampling configuration: an HTTP client for OpenAI-compatible, Ollama or custom gateways, and a batched Hugging Face generator for open-weight checkpoints on the university's Slurm cluster (NVIDIA A100 80 GB, Apptainer containers). Runs are resumable, and checkpoints are rejected if any weights fail to load.
- **Store.** SQLite with stories, every annotation round (with the taxonomy version it used), classifier predictions and judge scores; everything can be exported and re-imported as JSONL.
- **Annotation.** A web interface for human annotators (multi-label, span highlighting, keyboard shortcuts) and a batch protocol for LLM annotators. Protocol v2.1 defines casting by explicit criteria: gender by jobs (C1), rank (C2) and household tasks (C3), and ethnicity or nationality by status (C4), each with reference lists and

exclusions; everything else is judged as substantive.

- **Classifier.** One logistic regression per label over word TF-IDF and 79 stereotype co-occurrence features (gender cues near job and trait words, out-group terms near negative or positive words), semi-supervised by self-training on unlabelled stories, with thresholds tuned on out-of-fold predictions and a lexical prior that fades as labels accumulate. Optional LLM-judge scores are stacked as extra features. Each flag comes with its evidence sentences and top features.
- **Analyzers and report.** A default-demographics module (protagonist gender, name-based origin, name diversity) and the lexical analyzer feed a report with per-model rates, 95% Wilson intervals, per-category and per-premise tables (Markdown, JSON and a web page).

5 Data and processing

Data. All text is generated for the study: there is no scraped or personal data. The model set spans two groups: proprietary Claude models (Haiku, Sonnet, Opus) and open-weight models run on the cluster (Gemma 4 E4B, Qwen3.5 4B, 9B and 27B), to be extended with further open families (Llama, Mistral, OLMo, gpt-oss). Each model writes three stories per premise (267 per model); 1,380 stories have been generated so far. Labels come from blind annotators following the written protocol; LLM annotators stand in for human readers until people spot-check their labels.

Processing. Each step is scripted and can be rerun:

- **Prompting.** Premise plus a fixed instruction (about 250 words, every character named, no title or commentary) and a fixed system prompt that asks only for good fiction.
- **Generation.** Batched sampling with temperature 0.9, top-p 0.95, no top-k and at most 700 new tokens, thinking mode off; samples of each premise are generated in rounds so a cut-short run stays balanced.
- **Cleaning.** Strip reasoning blocks, titles and code fences; drop outputs under 20 words; deduplicate on a hash of model, premise and text; keep the raw output for auditing.
- **Sampling for annotation.** One story per premise per model; batches are shuffled under anonymous ids and every batch mixes all models, so annotator leniency cannot favour one family.
- **Validation.** Labels must belong to the taxonomy, and every label needs an exact quote that is located in the story and stored as a span; anchors from earlier rounds measure drift.
- **Training and evaluation.** 5-fold cross-validation over labelled stories (unlabelled stories stay on the training side); per-label precision, recall and F1, macro and micro F1, and bias-versus-none accuracy.
- **Reporting.** Any-label and beyond-casting rates per model with Wilson intervals, two-proportion tests between families, per-category breakdowns and inter-annotator agreement.

Next milestones. M2: a balanced annotation round under protocol v2.1 across all models and two to three additional open-weight families. M3: an LLM judge from a family that is not under test, the classifier with judge features, and a human spot-check of the LLM labels. M4: final analysis and report. **Limitations** to keep in view: name-based origin is a coarse signal, and LLM annotators are a stand-in for human readers.

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